

R Basic Course

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About Course__R

Description and educational material for introductory course on R, descriptive statistics, and visualizations with applications in fisheries.

This course has these sections;

- 1. **useR**. Intro to R and basic features.
- 2. **Exploratory Data Analysis**. Basic concepts for data exploration.
- 3. **Descriptive Statistics**. Essential statistical calculations.
- 4. **Visualization**. Graphics and Maps using modern packages
- 5. **Efficient code**. Organization strategies.
- 6. **Automatization and sharing** Report generation and result sharing.
- 7. **Management of Bibliographic** sources and additional materials.

All codes and used in this course can be founded in this repository with the structure to do this Ebook [Repo R Course](#)

Chapter 1

useR. Intro to R and basic features

Source in this [link](#)

[An introduction to R](#)

Chapter 2

Exploratory Data Analysis

Exploratory Data Analysis (EDA) is a crucial first step in analyzing data. It provides a summary of the main characteristics of datasets and helps identify patterns, outliers, and relationships.

libraries

```
library(tidyverse)
```

2.1 Summary Statistics in R

One of the simplest ways to start analyzing data is to generate summary statistics.

```
# Simple summary of built-in dataset  
summary(mtcars)
```

##	mpg	cyl	disp	hp
##	Min. :10.40	Min. :4.000	Min. : 71.1	Min. : 52.0
##	1st Qu.:15.43	1st Qu.:4.000	1st Qu.:120.8	1st Qu.: 96.5
##	Median :19.20	Median :6.000	Median :196.3	Median :123.0
##	Mean :20.09	Mean :6.188	Mean :230.7	Mean :146.7
##	3rd Qu.:22.80	3rd Qu.:8.000	3rd Qu.:326.0	3rd Qu.:180.0
##	Max. :33.90	Max. :8.000	Max. :472.0	Max. :335.0
##	drat	wt	qsec	vs
##	Min. :2.760	Min. :1.513	Min. :14.50	Min. :0.0000
##	1st Qu.:3.080	1st Qu.:2.581	1st Qu.:16.89	1st Qu.:0.0000
##	Median :3.695	Median :3.325	Median :17.71	Median :0.0000
##	Mean :3.597	Mean :3.217	Mean :17.85	Mean :0.4375
##	3rd Qu.:3.920	3rd Qu.:3.610	3rd Qu.:18.90	3rd Qu.:1.0000
##	Max. :4.930	Max. :5.424	Max. :22.90	Max. :1.0000

```
##           am           gear           carb
## Min.      :0.0000   Min.      :3.000   Min.      :1.000
## 1st Qu.:0.0000   1st Qu.:3.000   1st Qu.:2.000
## Median :0.0000   Median :4.000   Median :2.000
## Mean      :0.4062   Mean      :3.688   Mean      :2.812
## 3rd Qu.:1.0000   3rd Qu.:4.000   3rd Qu.:4.000
## Max.      :1.0000   Max.      :5.000   Max.      :8.000
```

2.2 Data Handling

2.2.1 Loading and Inspecting Data

EDA often begins by loading the dataset and inspecting its structure.

```
# Load CSV data
url <- "https://people.sc.fsu.edu/~jburkardt/data/csv/hw_200.csv"
data <- read.csv(url)

# View first rows of the data
head(data)
```

```
##      Index
## 1       2
## 2       3
## 3       4
## 4       5
## 5       6
## 6       7
## Height.Inches...Weight.Pounds..1..65.78..112.99.2..71.52..136.49.3..69.40..153.03
## 1
## 2
## 3
## 4
## 5
## 6
## Weight.Pounds.
## 1      136.49
## 2      153.03
## 3      142.34
## 4      144.30
## 5      123.30
## 6      141.49
```

2.3 Data Manipulation

2.3.1 Subsetting Data

You can select specific columns or rows based on logical operations.

```
# Select rows where mpg is greater than
# 20
subset_mpg <- mtcars[mtcars$mpg > 20, ]
head(subset_mpg)
```

```
##           mpg cyl  disp  hp drat   wt  qsec vs am gear carb
## Mazda RX4      21.0   6 160.0 110 3.90 2.620 16.46 0  1   4    4
## Mazda RX4 Wag  21.0   6 160.0 110 3.90 2.875 17.02 0  1   4    4
## Datsun 710     22.8   4 108.0  93 3.85 2.320 18.61 1  1   4    1
## Hornet 4 Drive 21.4   6 258.0 110 3.08 3.215 19.44 1  0   3    1
## Merc 240D      24.4   4 146.7  62 3.69 3.190 20.00 1  0   4    2
## Merc 230       22.8   4 140.8  95 3.92 3.150 22.90 1  0   4    2
```

2.3.2 Using dplyr for Data Manipulation

The dplyr package makes data manipulation efficient and readable.

```
# Load the dplyr package
library(dplyr)

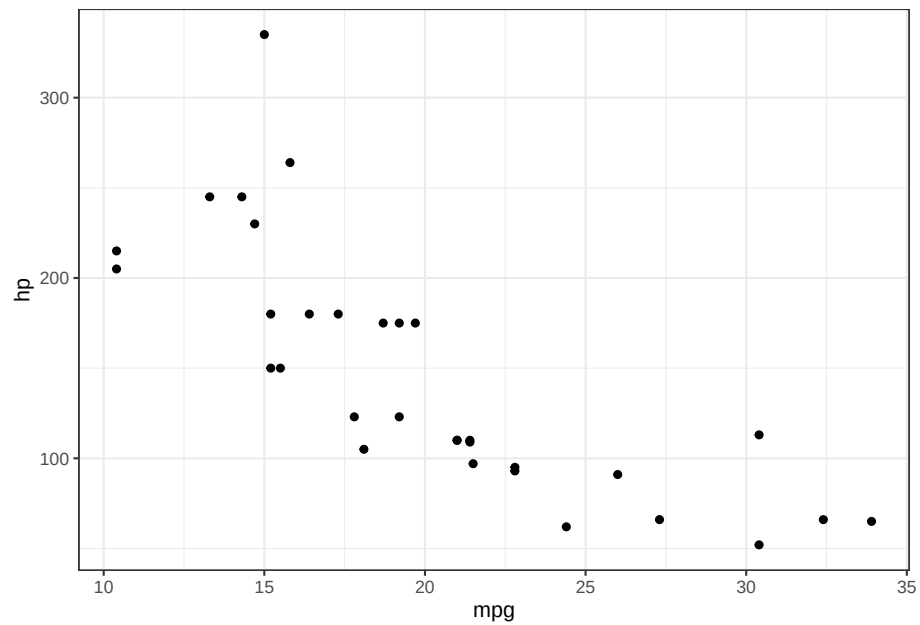
# Filter and summarize data
mtcars %>%
  filter(mpg > 20) %>%
  summarise(avg_hp = mean(hp), count = n())
```

```
##   avg_hp count
## 1   88.5    14
```

2.4 Visualization with Base R

Visualizations are essential for gaining insights quickly.

```
# Simple scatter plot
carsplot <- ggplot() + geom_point(data = mtcars,
  aes(mpg, hp)) + theme_bw()
carsplot
```



2.5 Final Thoughts

- EDA is critical for understanding your data.
- Data handling ensures clean datasets.
- Data manipulation prepares data for in-depth analysis.
- Visualization helps reveal insights quickly.

2.6 References

[Basic R](#)

[Data Explorer](#)

[EMSE 4572 CleanData](#)

Chapter 3

Descriptive Statistics.

Descriptive statistics summarize and organize data. They help us understand patterns and key characteristics in a dataset.

We'll explore basic concepts using the `penguins` dataset from the `palmerpenguins` package.

3.1 The penguins Dataset

```
library(tidyverse)
library(palmerpenguins)
library(kableExtra)
data("penguins")
kbl(head(penguins, n = 4))
```

species	island	bill_length_mm	bill_depth_mm	flipper_length_mm	body_mass_g	sex	year
Adelie	Torgersen	39.1	18.7	181	3750	male	2007
Adelie	Torgersen	39.5	17.4	186	3800	female	2007
Adelie	Torgersen	40.3	18.0	195	3250	female	2007
Adelie	Torgersen	NA	NA	NA	NA	NA	2007

3.2 Basic Data Exploration

```
# Check dimensions and structure
dim(penguins)
```

```
## [1] 344 8
```

```
str(penguins)
```

```
## tibble [344 x 8] (S3: tbl_df/tbl/data.frame)
## $ species      : Factor w/ 3 levels "Adelie","Chinstrap",...: 1 1 1 1 1 1 1 1 1 1 ...
## $ island       : Factor w/ 3 levels "Biscoe","Dream",...: 3 3 3 3 3 3 3 3 3 3 ...
## $ bill_length_mm : num [1:344] 39.1 39.5 40.3 NA 36.7 39.3 38.9 39.2 34.1 42 ...
## $ bill_depth_mm  : num [1:344] 18.7 17.4 18 NA 19.3 20.6 17.8 19.6 18.1 20.2 ...
## $ flipper_length_mm: int [1:344] 181 186 195 NA 193 190 181 195 193 190 ...
## $ body_mass_g    : int [1:344] 3750 3800 3250 NA 3450 3650 3625 4675 3475 4250 ...
## $ sex           : Factor w/ 2 levels "female","male": 2 1 1 NA 1 2 1 2 NA NA ...
## $ year          : int [1:344] 2007 2007 2007 2007 2007 2007 2007 2007 2007 2007
```

3.3 Summary Statistics

```
# Quick statistical overview
summary(penguins)
```

```
##      species      island  bill_length_mm  bill_depth_mm
## Adelie   :152   Biscoe   :168   Min.      :32.10   Min.      :13.10
## Chinstrap: 68   Dream    :124   1st Qu.:39.23   1st Qu.:15.60
## Gentoo   :124   Torgersen: 52   Median :44.45   Median :17.30
##                                     Mean   :43.92   Mean   :17.15
##                                     3rd Qu.:48.50   3rd Qu.:18.70
##                                     Max.    :59.60   Max.    :21.50
##                                     NA's    :2      NA's    :2
## flipper_length_mm  body_mass_g      sex      year
## Min.      :172.0    Min.      :2700   female:165   Min.      :2007
## 1st Qu.:190.0    1st Qu.:3550   male  :168   1st Qu.:2007
## Median :197.0    Median :4050   NA's   : 11   Median :2008
## Mean   :200.9    Mean   :4202                   Mean   :2008
## 3rd Qu.:213.0    3rd Qu.:4750                   3rd Qu.:2009
## Max.    :231.0    Max.    :6300                   Max.    :2009
## NA's     :2      NA's     :2
```

3.4 Central Tendency

- **Mean:** average value
- **Median:** middle value
- **Mode:** most frequent value

```
mean(penguins$bill_length_mm, na.rm = TRUE)
```

```
## [1] 43.92193
```

```
median(penguins$bill_length_mm, na.rm = TRUE)
```

```
## [1] 44.45
```

3.5 Variability

- **Range:** difference between max and min
- **Variance:** spread of the data
- **Standard deviation:** square root of variance

```
range(penguins$bill_length_mm, na.rm = TRUE)
```

```
## [1] 32.1 59.6
```

```
var(penguins$bill_length_mm, na.rm = TRUE)
```

```
## [1] 29.80705
```

```
sd(penguins$bill_length_mm, na.rm = TRUE)
```

```
## [1] 5.459584
```

3.6 Grouped Summaries

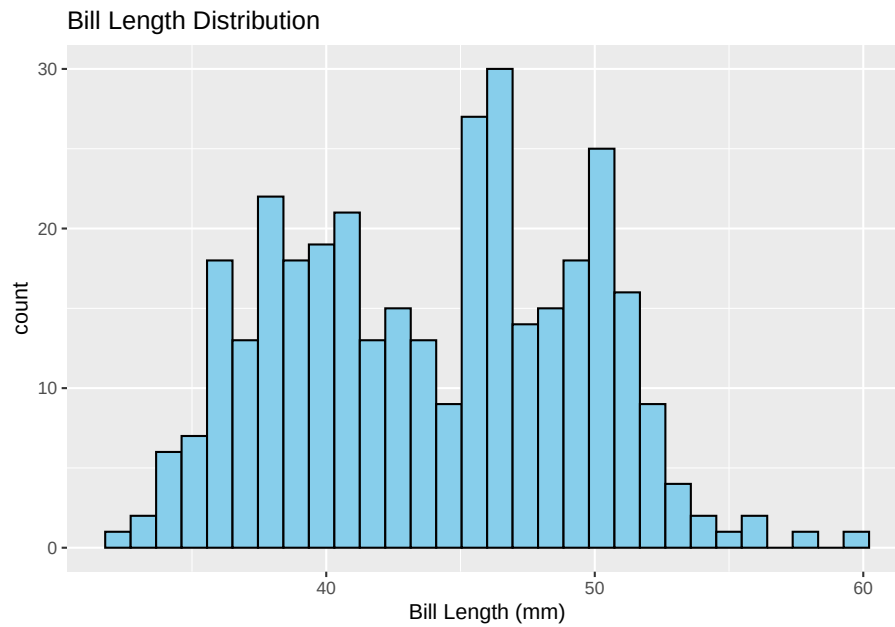
```
# Grouped mean by species  
library(dplyr)  
penguins %>%  
  group_by(species) %>%  
  summarise(mean_bill_length = mean(bill_length_mm,  
    na.rm = TRUE))
```

```
## # A tibble: 3 x 2  
##   species mean_bill_length  
##   <fct>         <dbl>  
## 1 Adelie         38.8  
## 2 Chinstrap      48.8  
## 3 Gentoo         47.5
```

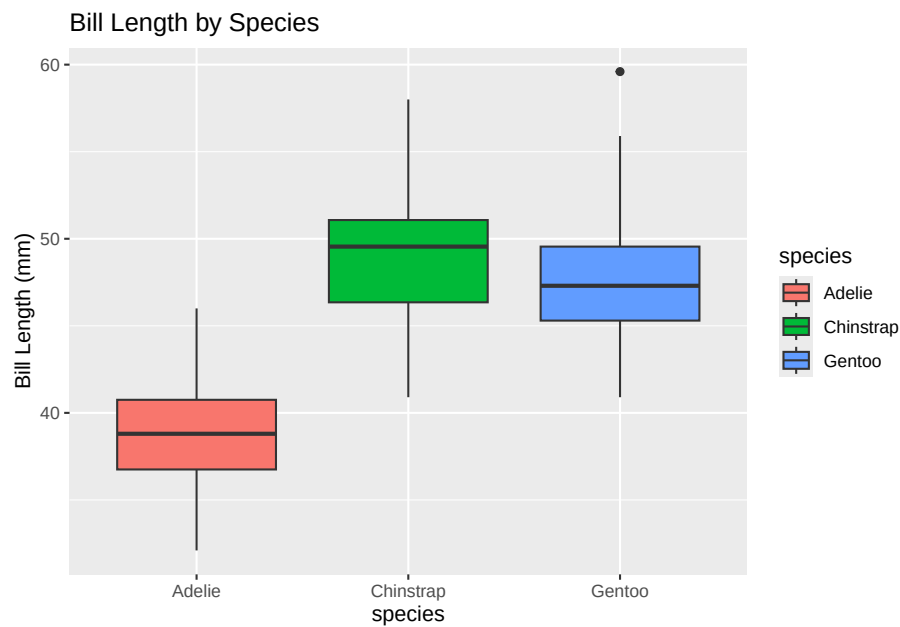
3.7 Visualization: Histograms

```
library(ggplot2)  
ggplot(penguins, aes(x = bill_length_mm)) +  
  geom_histogram(bins = 30, fill = "skyblue",
```

```
color = "black") + labs(title = "Bill Length Distribution",  
x = "Bill Length (mm)")
```



```
ggplot(penguins, aes(x = species, y = bill_length_mm,  
fill = species)) + geom_boxplot() + labs(title = "Bill Length by Species",  
y = "Bill Length (mm)")
```



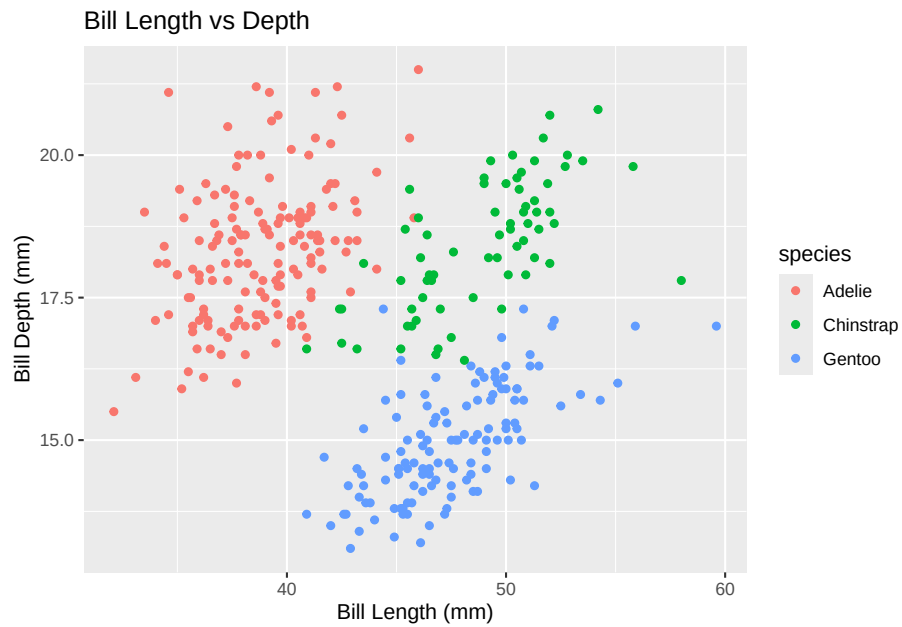
3.9 Correlation

```
cor(penguins$bill_length_mm, penguins$bill_depth_mm,
    use = "complete.obs")
```

```
## [1] -0.2350529
```

3.10 Scatter Plot for Relationship

```
ggplot(penguins, aes(x = bill_length_mm,
    y = bill_depth_mm, color = species)) +
  geom_point() + labs(title = "Bill Length vs Depth",
    x = "Bill Length (mm)", y = "Bill Depth (mm)")
```



3.11 Conclusion

- Descriptive statistics are fundamental for understanding data.
- R provides powerful tools to calculate, visualize, and interpret data patterns.

Questions?

Sources in this [link](#)

[here](#)

[Data Types & Probability Distributions](#)

Chapter 4

Visualization.

Graphics and Maps using modern packages

4.1 Graphics

Footnotes are put inside the square brackets after a caret `^[]`. Like this one ¹.

4.2 Simple maps

Reference items in your bibliography file(s) using `@key`.

For example, we are using the **bookdown** package [Xie, 2024] (check out the last code chunk in `index.Rmd` to see how this citation key was added) in this sample book, which was built on top of R Markdown and **knitr** [Xie, 2015] (this citation was added manually in an external file `book.bib`). Note that the `.bib` files need to be listed in the `index.Rmd` with the YAML `bibliography` key.

The RStudio Visual Markdown Editor can also make it easier to insert citations: <https://rstudio.github.io/visual-markdown-editing/#/citations>

¹This is a footnote.

Chapter 5

Efficient Code

5.1 Equations

Here is an equation.

$$f(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad (5.1)$$

You may refer to using `\@ref{eq:binom}`, like see Equation (5.1).

Read more here <https://bookdown.org/yihui/bookdown/markdown-extensions-by-bookdown.html>.

5.2 Callout blocks

The R Markdown Cookbook provides more help on how to use custom blocks to design your own callouts: <https://bookdown.org/yihui/rmarkdown-cookbook/custom-blocks.html>

Chapter 6

Reporting and Sharing

6.1 Publishing

HTML books can be published online, see: <https://bookdown.org/yihui/bookdown/publishing.html>

6.2 404 pages

By default, users will be directed to a 404 page if they try to access a webpage that cannot be found. If you'd like to customize your 404 page instead of using the default, you may add either a `_404.Rmd` or `_404.md` file to your project root and use code and/or Markdown syntax.

6.3 Metadata for sharing

Bookdown HTML books will provide HTML metadata for social sharing on platforms like Twitter, Facebook, and LinkedIn, using information you provide in the `index.Rmd` YAML. To setup, set the `url` for your book and the path to your `cover-image` file. Your book's `title` and `description` are also used.

This `gitbook` uses the same social sharing data across all chapters in your book—all links shared will look the same.

Specify your book's source repository on GitHub using the `edit` key under the configuration options in the `_output.yml` file, which allows users to suggest an edit by linking to a chapter's source file.

Read more about the features of this output format here:

<https://pkgs.rstudio.com/bookdown/reference/gitbook.html>

Or use:

```
`?`(bookdown::gitbook)
```

Bibliography

Yihui Xie. *Dynamic Documents with R and knitr*. Chapman and Hall/CRC, Boca Raton, Florida, 2nd edition, 2015. URL <http://yihui.org/knitr/>. ISBN 978-1498716963.

Yihui Xie. *bookdown: Authoring Books and Technical Documents with R Markdown*, 2024. URL <https://github.com/rstudio/bookdown>. R package version 0.40.